

SplitFed Speech Emotion Classification

Comparative Experiment Tables

Summary tables

Table 1: Train/test split statistics for the E0 cross-corpus setup.

Dataset	Split	Samples	Actors	Anger	Non-anger	Anger share
CREMA-D	Train	5890	72	1006	4884	17.08%
CREMA-D	Test	1552	19	265	1287	17.07%
RAVDESS	Train	1140	19	152	988	13.33%
RAVDESS	Test	300	5	40	260	13.33%
SAVEE	Train	360	3	45	315	12.50%
SAVEE	Test	120	1	15	105	12.50%

Table 2: Aggregated dataset-level statistics for the E0 setup.

Dataset	Samples	Actors	Anger	Non-anger	Mean dur. (s)	Std. dur. (s)
CREMA-D	7442	91	1271	6171	2.5429	0.5060
RAVDESS	1440	24	192	1248	3.7007	0.3367
SAVEE	480	4	60	420	3.8395	1.0780

Table 3: Pairwise Maximum Mean Discrepancy (MMD) across datasets without normalization.

Dataset	CREMA-D	RAVDESS	SAVEE
CREMA-D	-	0.381637	0.389779
RAVDESS	0.381637	-	0.332834
SAVEE	0.389779	0.332834	-

Table 4: Pairwise Maximum Mean Discrepancy (MMD) across datasets after normalization.

Dataset	CREMA-D	RAVDESS	SAVEE
CREMA-D	-	0.009802	0.018052
RAVDESS	0.009802	-	0.017652
SAVEE	0.018052	0.017652	-

Table 5: E1 local-corpus evaluation at seed 42: anger F1 on each cross-corpus test set. Every model is trained only on the corpus named in its row, using the actor-disjoint split from the rerun.

Train model	CREMA-D test	RAVDESS test	SAVEE test
CREMA-D	0.6922	0.1395	0.6111
RAVDESS	0.2778	0.6207	0.4103
SAVEE	0.3846	0.2621	0.7027

Table 6: E2 centralized versus FedAvg quality at seed 42. Metrics are sample-weighted across the three corpus-specific test sets.

Method	Accuracy	Anger F1	Macro F1	Precision	Recall	PR-AUC
Centralized	0.8920	0.6864	0.8102	0.6655	0.7409	0.7772
FedAvg	0.8966	0.6598	0.7990	0.7101	0.6645	0.7757

Table 7: E3 protocol comparison at round 50 of the seed-42 rerun. Metrics are sample-weighted across the three corpus-specific test sets. The protocols retain their native workload and server-model ownership semantics.

Method	Accuracy	Anger F1	Macro F1	Precision	Recall	PR-AUC
MergeSFL	0.8763	0.6330	0.7786	0.6632	0.6845	0.7290
SFLv1	0.8844	0.6720	0.8005	0.6523	0.7298	0.7622
SFLour	0.8727	0.6498	0.7854	0.6259	0.7313	0.7691

Table 8: E4 SFLour workload-policy quality at round 50 of the seed-42 rerun. Metrics are sample-weighted across the three corpus-specific test sets.

Policy	Accuracy	Anger F1	Macro F1	Precision	Recall	PR-AUC
Full Local Epoch	0.8727	0.6498	0.7854	0.6259	0.7313	0.7691
Fixed Maximum Steps	0.8925	0.6601	0.7980	0.7040	0.6426	0.7292
Balanced Workload	0.8595	0.6265	0.7696	0.5723	0.7206	0.7113

Table 9: E4 SFLour workload-policy cost at round 50 of the seed-42 rerun. Batches/round are listed as CREMA-D/RAVDESS/SAVEE. The fixed budget of 143 is the RAVDESS (middle client) epoch length at batch size 8. GB and MB use decimal units.

Policy	Batches/round	Wall time (s)	CPU user (s)	CPU sys (s)	Transport (GB)	Messages	Peak RSS (GB)
Full Local Epoch	737/143/45	801.30	1629.39	311.44	24.555	95150	1.937
Fixed Maximum Steps	143/143/143	295.22	825.63	129.71	11.566	45550	1.915
Balanced Workload	64/64/45	179.91	343.26	59.61	4.865	19950	1.952

Table 10: Resource usage for the E2–E4 seed-42 rerun. Transport volume uses decimal GB, peak resident memory uses decimal GB, and peak CUDA allocation uses decimal MB. Process CPU times are summed across roles.

Method	Wall time (s)	CPU user (s)	CPU sys (s)	Transport (GB)	Messages	Peak RSS (GB)	Peak CUDA (MB)
E2 Centralized	494.21	358.08	84.79	0.000	0	2.165	156.6
E2 FedAvg	553.38	2760.32	153.73	2.103	300	2.151	156.6
E3 MergeSFL	251.53	417.13	90.71	7.860	31520	1.911	185.8
E3 SFLv1	807.99	1681.41	318.75	24.555	95300	1.938	205.2
E3 SFLour	801.30	1629.39	311.44	24.555	95150	1.937	250.7
E4 Balanced	179.91	343.26	59.61	4.865	19950	1.952	250.7
E4 Fixed steps	295.22	825.63	129.71	11.566	45550	1.915	250.7

Table 11: E1 local cross-corpus anger F1 over seeds 42–47. Values are mean $\pm$ sample standard deviation across six runs. Each seed produces a new actor-disjoint split.

Train corpus	CREMA-D test	RAVDESS test	SAVEE test
CREMA-D	0.695 ± 0.015	0.040 ± 0.055	0.334 ± 0.253
RAVDESS	0.344 ± 0.146	0.651 ± 0.139	0.264 ± 0.146
SAVEE	0.312 ± 0.064	0.192 ± 0.098	0.488 ± 0.255

Table 12: E2 centralized and FedAvg anger F1 over seeds 42–47. Values are mean $\pm$ sample standard deviation across six actor-disjoint splits.

Method	CREMA-D	RAVDESS	SAVEE
Centralized	0.696 ± 0.032	0.649 ± 0.077	0.270 ± 0.246
FedAvg	0.727 ± 0.028	0.353 ± 0.146	0.338 ± 0.215

Table 13: E3 and E4 quality and runtime over seeds 42–47. Quality columns are macro-averaged across corpora except for weighted anger F1. Values are mean $\pm$ sample standard deviation across six runs. Runtime is wall-clock time in seconds.

Method	Accuracy	Anger F1	Macro F1	UAR	PR-AUC	Weighted anger F1	Time (s)
MergeSFL	0.820 ± 0.123	0.519 ± 0.080	0.692 ± 0.092	0.746 ± 0.027	0.662 ± 0.086	0.601 ± 0.050	262.2 ± 8.5
SFLv1	0.842 ± 0.083	0.532 ± 0.105	0.711 ± 0.081	0.740 ± 0.053	0.683 ± 0.068	0.643 ± 0.032	837.9 ± 19.3
SFLour full epoch	0.837 ± 0.093	0.535 ± 0.076	0.709 ± 0.062	0.732 ± 0.045	0.676 ± 0.066	0.640 ± 0.026	832.2 ± 15.9
SFLour balanced	0.808 ± 0.111	0.551 ± 0.063	0.701 ± 0.080	0.758 ± 0.046	0.644 ± 0.059	0.620 ± 0.021	187.2 ± 4.7
SFLour fixed steps	0.842 ± 0.086	0.560 ± 0.071	0.725 ± 0.063	0.752 ± 0.046	0.695 ± 0.054	0.636 ± 0.021	316.6 ± 17.0